

TrES-1b

R-1313

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_190522 · created 2026-07-31T18:48:17+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458625.7793 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.003 +/- 0.053
Transit depth (Rp/Rs)^2	0.00099 +/- 0.034 [%]
Orbital Inclination (inc)	89.42 +/- 1.44
Airmass coefficient 1 (a1)	2.63 +/- 1.37
Airmass coefficient 2 (a2)	-0.44 +/- 0.2
Scatter in the residuals of the lightcurve fit is	54.87 %
Best Comparison Star	None
Optimal Aperture	2.64
Optimal Annulus	5.0
Transit Duration (day)	0.0917 +/- 0.0068

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.003 ± 0.053 vs 0.1358 (deviation 1.71σ)
Duration fit vs published	0.0917 d vs 0.1042 d (deviation 1.25σ)
Mid-time O–C	-4.7 min
Depth SNR	0.0
Residual scatter	54.87 %

Light curve

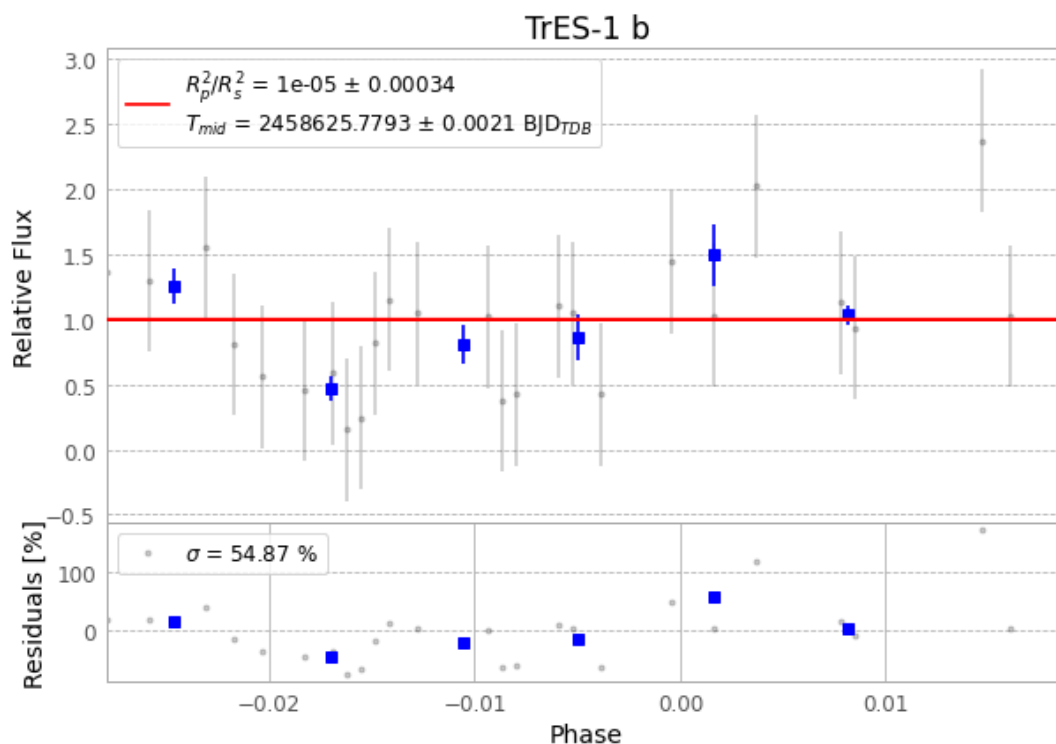


Figure 1 — FinalLightCurve_TrES-1 b_2019-05-21.png (SHA-256 f3852e7a1db7810e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [576, 420], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1c262c9badfd913a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

91 FITS frames (60 MB), TRES-1190522043312.FITS → TRES-1190522090312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-190522.log (fcae541c01a2...), FinalLightCurve_TrES-1 b_2019-05-21.png (f3852e7a1db7...), FinalLightCurve_TrES-1 b_2019-05-21.csv (5623613dc5bb...)

Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 18:48Z.